SUBJECT: ASTRONOMY / PLANETARY SCIENCE

# Mars Planetary Science & Autonomous Exploration Missions

*Prepared for Offline AI Knowledge Ingestion & RAG Verification*

## 1. Martian Atmospheric & Geological Characteristics

Mars possesses a thin atmosphere composed of 95.3% carbon dioxide, 2.6% nitrogen, and 1.9% argon, with an average surface atmospheric pressure of 610 Pascals (less than 1% of Earth's sea-level pressure). Surface temperatures range from -125°C at the winter poles to 20°C during equatorial summer.

## 2. Surface Exploration & Biosignature Detection

- **Perseverance Rover:** Equipped with the SHERLOC ultraviolet Raman spectrometer and PIXL X-ray fluorescence system to identify organic molecules and biosignatures in Jezero Crater.
- **MOXIE Experiment:** Successfully produced breathable oxygen (O■) on Mars by solid oxide electrolysis of atmospheric carbon dioxide at 800°C.
- **Sample Return Architecture:** Joint NASA/ESA mission profile utilizing an Earth Return Orbiter and Mars Ascent Vehicle to return hermetically sealed core samples to Earth.